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Math 102 Final Exam Term 252

Original exam date: 13 May 2026. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. $\int \frac{\cos\theta}{\sqrt{1-\sin\theta}}\,d\theta =$

(a) $\ln\left(\sqrt{1-\sin\theta}\right)+C$

(b) $2\sqrt{1-\sin\theta}+C$

(c) $\sqrt{1-\sin\theta}+C$

(d) $-\sqrt{1-\sin\theta}+C$

(e) $-2\sqrt{1-\sin\theta}+C$

2. Which one of the following series is/are divergent?

$$\text{(I)}\ \sum_{n=1}^{\infty}\sin\frac{1}{n} \qquad \text{(II)}\ \sum_{n=1}^{\infty}e^{-n^2} \qquad \text{(III)}\ \sum_{n=1}^{\infty}\frac{5}{4^n+1}$$

(a) II only

(b) II and III only

(c) I and II only

(d) I only

(e) I and III only

3. Evaluate

$$\int_1^{\infty}\frac{\ln x}{x}\,dx.$$

(a) converges to 1

(b) diverges

(c) converges to 0

(d) converges to e^2

(e) converges to e

4. The coefficient of x^7 in the Maclaurin series for the function $f(x) = \sinh x$ is:

(a) $\dfrac{-1}{7!}$

(b) $\dfrac{4}{7!}$

(c) $\dfrac{3}{7!}$

(d) $\dfrac{2}{7!}$

(e) $\dfrac{1}{7!}$

5. The series $\sum_{n=0}^{\infty} \frac{(-1)^n\, n!\, (x-5)^n}{3^n}$ converges

(a) on the interval $(-5, 5]$

(b) on the interval $[-5, 5]$

(c) only for $x = 5$

(d) on the interval $(-\infty, \infty)$

(e) on the interval $(-5, 5)$

6. Using the Remainder Theorem, the least number of terms required to approximate the sum of the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ with an error of less than 0.02 is:

(a) 4

(b) 3

(c) 9

(d) 10

(e) 7

7. The volume of the solid generated by revolving the region bounded by the graphs of $y = e^{-x}$, $y = 0$, $x = 0$, and $x = 1$ about the x-axis is:

(a) $\dfrac{\pi}{2}\left(1 - \dfrac{1}{e}\right)$

(b) $\dfrac{\pi}{2}\left(1 + \dfrac{1}{e^2}\right)$

(c) $\pi\left(1 - \dfrac{1}{e}\right)$

(d) $\dfrac{\pi}{2}\left(1 + \dfrac{1}{e}\right)$

(e) $\dfrac{\pi}{2}\left(1 - \dfrac{1}{e^2}\right)$

8. Evaluate

$$\int_3^6 \frac{dx}{\sqrt{36-x^2}}.$$

(a) converges to 0

(b) converges to $\frac{\pi}{2}$

(c) converges to $\frac{\pi}{3}$

(d) diverges

(e) converges to $\frac{\pi}{6}$

9. Which one of the following series is/are convergent?

$$\text{(I)} \sum_{n=1}^{\infty} \frac{n}{n^4+2n^2+1} \qquad \text{(II)} \sum_{n=1}^{\infty} \frac{1}{n^{1.001}} \qquad \text{(III)} \sum_{n=1}^{\infty} \frac{1}{n^{0.9}}$$

(a) III only

(b) I, II and III

(c) I and III only

(d) I and II only

(e) II and III only

10. A power series representation for the function $f(x) = \dfrac{3x-1}{x^2-1}$ centered at $c = 0$ is:

(a) $\sum_{n=0}^{\infty} [2(-1)^n + 1]\, x^n,\ (-1, 1)$

(b) $\sum_{n=0}^{\infty} [2(-1)^n + 2]\, x^n,\ (-1, 1)$

(c) $\sum_{n=0}^{\infty} [(-1)^n - 1]\, x^n,\ (-1, 1)$

(d) $\sum_{n=0}^{\infty} [2(-1)^n - 1]\, x^n,\ (-1, 1)$

(e) $\sum_{n=0}^{\infty} [(-1)^n + 1]\, x^n,\ (-1, 1)$

11. The sequence whose nth term is $a_n = \dfrac{\cos 2n}{3^n}$ converges to:

(a) $\frac{3}{2}$

(b) 3

(c) 0

(d) $\frac{2}{3}$

(e) 2

12. $\int x \sec^2 x \, dx =$

(a) $\tan x + \ln|\cos x| + C$

(b) $x \tan x + \ln|\cos x| + C$

(c) $x^2 \tan x + \ln|\cos x| + C$

(d) $x \tan x - \ln|\cos x| + C$

(e) $x \sec x + \ln|\cos x| + C$

13. The series $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$

(a) converges by the Integral Test

(b) converges absolutely

(c) converges conditionally

(d) diverges

(e) diverges by the Alternating Series Test

14. If a Taylor polynomial for $f(x) = \sqrt{x}$ centered at $c = 4$ is written in the form $p_2(x) = a_2x^2 + a_1x + a_0$, then $64a_2 + 8a_1 + 4a_0 =$

(a) 8

(b) 4

(c) 7

(d) 5

(e) 6

15. Find the sum

$$\sum_{n=0}^{\infty} \left(-\frac{1}{5}\right)^n + \sum_{n=1}^{\infty} \frac{1}{(2n+1)(2n+3)}.$$

(a) 1

(b) 3

(c) 5

(d) 4

(e) 2

16. $\int \frac{\tan^3 x}{\sqrt{\sec x}}\,dx =$

(a) $\frac{2}{3}(\sec x)^{3/2} - (\sec x)^{-1/2} + C$

(b) $\frac{2}{3}(\sec x)^{3/2} - 2(\sec x)^{-1/2} + C$

(c) $\frac{1}{3}(\sec x)^{3/2} + (\sec x)^{-1/2} + C$

(d) $\frac{2}{3}(\sec x)^{3/2} + 2(\sec x)^{-1/2} + C$

(e) $\frac{1}{2}(\sec x)^{3/2} + (\sec x)^{-1/2} + C$

17. $\int \frac{4}{x^2\sqrt{4+x^2}}\,dx =$

(a) $\frac{\sqrt{4+x^2}}{2} + C$

(b) $\frac{\sqrt{4+x^2}}{x} + C$

(c) $\frac{x\sqrt{4+x^2}}{2} + C$

(d) $\frac{-\sqrt{4+x^2}}{x} + C$

(e) $\frac{-\sqrt{4+x^2}}{2} + C$

18. The arc length of the graph of the function $f(x) = \frac{1}{3}x^{3/2} - 1$ over the interval $[2, 6]$ is:

(a) $\frac{1}{4}\left(10^{3/2} - 6^{3/2}\right)$

(b) $\frac{1}{2}\left(10^{3/2} + 6^{3/2}\right)$

(c) $\frac{1}{3}\left(10^{3/2} + 6^{3/2}\right)$

(d) $\frac{1}{3}\left(10^{3/2} - 6^{3/2}\right)$

(e) $\frac{1}{2}\left(10^{3/2} - 6^{3/2}\right)$

19. Which one of the following series is/are divergent?

$$\text{(I)} \sum_{n=0}^{\infty} \frac{6^n}{(n+1)^3} \qquad \text{(II)} \sum_{n=0}^{\infty} \frac{(n!)^2}{(3n)!} \qquad \text{(III)} \sum_{n=1}^{\infty} \left(\frac{-3n}{2n+1} \right)^{3n}$$

(a) I and III only

(b) I and II only

(c) II and III only

(d) I, II and III

(e) III only

20. The area of the region bounded by the graphs of the equations $y = 12 - x^2$, $y = x$, $x = -1$ and $x = 1$ is:

(a) $\frac{62}{3}$

(b) $\frac{70}{3}$

(c) $\frac{64}{3}$

(d) $\frac{65}{5}$

(e) $\frac{61}{3}$

Answer Key

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Q	Ans	Brief explanation
1	**(e)**	$u = 1 - \sin\theta$: $-\int u^{-1/2} du = -2\sqrt{1-\sin\theta} + C$.
2	**(d)**	I diverges (limit comparison with $\sum 1/n$); II and III converge (geometric bounds).
3	**(b)**	$\int_1^b = \frac{(\ln b)^2}{2} \to \infty$: diverges.
4	**(e)**	$\sinh x = \sum \frac{x^{2n+1}}{(2n+1)!}$, so the x^7 coefficient is $\frac{1}{7!}$.
5	**(c)**	Ratio gives $\frac{(n+1)\lvert x-5\rvert}{3} \to \infty$ for $x \neq 5$: only $x = 5$.
6	**(e)**	$\frac{1}{(N+1)^2} < 0.02 \Rightarrow (N+1)^2 > 50 \Rightarrow N = 7$.
7	**(e)**	Disks: $\pi \int_0^1 e^{-2x} dx = \frac{\pi}{2}(1 - e^{-2})$.
8	**(c)**	$\arcsin \frac{x}{6}\Big\rvert_3^6 = \frac{\pi}{2} - \frac{\pi}{6} = \frac{\pi}{3}$.
9	**(d)**	I $\sim 1/n^3$ converges, II ($p = 1.001$) converges, III ($p = 0.9$) diverges.
10	**(d)**	$\frac{1}{x-1} + \frac{2}{x+1} = \sum[2(-1)^n - 1]x^n$ on $(-1, 1)$.
11	**(c)**	$\lvert a_n\rvert \leq (1/3)^n \to 0$, so by squeeze the limit is 0.
12	**(b)**	Parts with $u = x$, $dv = \sec^2 x\, dx$: $x\tan x + \ln\lvert\cos x\rvert + C$.
13	**(c)**	Alternating series test passes, but $\sum n^{-1/2}$ diverges: conditional convergence.
14	**(d)**	$p_2 = -\frac{x^2}{64} + \frac{3x}{8} + \frac{3}{4}$, so $-1 + 3 + 3 = 5$.
15	**(a)**	Geometric $\frac{5}{6}$ plus telescoping $\frac{1}{6}$ gives 1.
16	**(d)**	$u = \sec x$: $\int (u^{1/2} - u^{-3/2}) du = \frac{2}{3}u^{3/2} + 2u^{-1/2} + C$.
17	**(d)**	$x = 2\tan\theta$: $\int \frac{\cos\theta}{\sin^2\theta} d\theta = -\csc\theta = -\frac{\sqrt{4+x^2}}{x} + C$.
18	**(d)**	$1 + (f')^2 = \frac{4+x}{4}$, $L = \int_2^6 \frac{\sqrt{4+x}}{2} dx = \frac{1}{3}(10^{3/2} - 6^{3/2})$.
19	**(a)**	Ratio: I gives $6 > 1$, II gives 0; root: III gives $27/8 > 1$. I and III diverge.
20	**(b)**	$\int_{-1}^1 (12 - x^2 - x) dx = 24 - \frac{2}{3} = \frac{70}{3}$.